



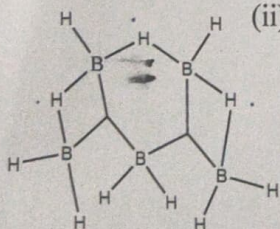
PART A (Answer ALL, Marks: 10 x 2 = 20)

1. What is calix[4]arene? Draw their four possible conformations.
2. Outline the preparation and structure of basic beryllium acetate.
3. Draw the structure of a) Beryllocene b) Borax
4. If BH_3 is a Lewis acid, how does it form adducts with CO and PF_3 ? Justify your explanation based on its bonding formation.
5. How many electrons are there in total in the diborane molecule, B_2H_6 ? In that, how many electrons are used for each possible bonding?
6. Discuss the bonding situation in Al_2Ph_6 , clearly showing the different bonding modes of the bridging C_6H_5^- anion ($\mu\text{-C}_6\text{H}_5^-$).
7. Briefly describe the structure and bonding of two allotropes of carbon.
8. Why is the existence of a silicon analogue of graphite unlikely?
9. Disilenes and other $\text{R}_2\text{M}=\text{MR}_2$ analogues of heavier group 14 elements are trans folded whereas alkenes are planar. Explain.
10. What are called polyphosphates and metaphosphates? Explain their difference with an example for each.

PART B (Answer ALL, Marks: 6 x 5 = 30)

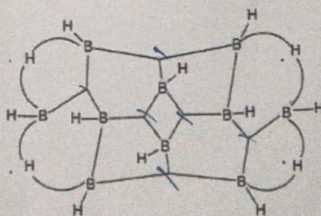
11. (a) What is 18-crown-6, and how is it named? Sketch its structure. (2 marks)
(b) What is [2.2.2]-cryptand? Sketch its structure (2 marks)
(c) The complexation (k_1) and release (k_{-1}) rates for 18-crown-6 reacting with KCl in H_2O are $4.8 \times 10^8 \text{ M}^{-1}\text{s}^{-1}$ and $3.7 \times 10^6 \text{ s}^{-1}$, respectively. The similar data for [2.2.2]-cryptand with KCl in H_2O is: $k_1 = 7.5 \times 10^6 \text{ M}^{-1}\text{s}^{-1}$ and $k_{-1} = 38 \text{ s}^{-1}$. Use these data to explain how cryptands are better complexing agents than crown ethers. (1 mark)
12. (a) Using the polyhedral electron pair approach (Wade's rule) predict the structures for the anions, B_2H_7^- and B_3H_8^- ? (1 mark)
(b) Using the polyhedral electron pair approach (Wade's rule) predict the structure of the basic framework of $\text{C}_2\text{B}_9\text{H}_{11}^{2-}$. (1 mark)
(c) Find the STYX number for the following polyhedral boranes and give their nomenclature. (3 marks)

(i)

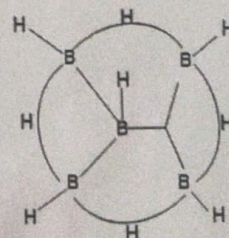


B

(ii)



(iii)



13. a) Draw the structure of compounds, $R_6B_3N_3$, $R_3B_3O_3$ and $R_6B_3P_3$. (1.5 marks)
- b) Which of these compounds will have the highest tendency to form a π -complex with $Cr(CO)_3$ and why? (1.5 marks)
- c) Borazine and cyclophosphazenes are isostructural with benzene; however, only borazine is considered as an organic compound whereas cyclophosphazenes are not considered. Justify your answer. (2 marks)

14. Find the products A-J in the below reactions:

- a) $CH_3MgCl + C_2H_5CN + H^+ \rightarrow A$
- b) $B_2O_3 (s) + 3H_2O (l) \rightarrow B$
- c) $B_2H_6 + NH_3 \rightarrow C$
- d) $AlCl_3$ with Al at 1270 K $\rightarrow D$
- e) $B_3N_3H_6 + 3HCl \rightarrow E$
- f) $Me_2SiCl_2 + 2H_2O \rightarrow F$
- g) $PCl_3 (s) + 3H_2O (l) \rightarrow G$
- h) $PCl_5 (s) + H_2O (l) \rightarrow H$
- i) $SO_3 (g) + H_2O (l) \rightarrow I$
- j) $Cl_2 (g) + 3F_2 (g) \rightarrow J$

15. a) How to define silicates with a general chemical formula. (1 mark)
- b) List the different types of silicates with their Si:O ratio (2 marks)
- c) Draw the structure of any two types of silicates and explain their bonding. (2 marks)
16. a) Arrange the interhalogens in their decreasing order of reactivity: (i) BrF_3 , IF_3 , and ClF_3 ; (ii) BrF , BrF_5 and BrF_3 . (2 marks)
- b) Use VSEPR to predict the shapes in the following interhalogens: (2 marks)
- (i) ICl_3 (ii) IF_5 (iii) $[ClF_2]^-$ (iv) $[ClF_4]^+$
- c) Draw the structure of the following polyhalogeno cations. (1 mark)
- (i) $[Br_3]^+$ (ii) $[I_5]^+$

Name:

Reg. No.:

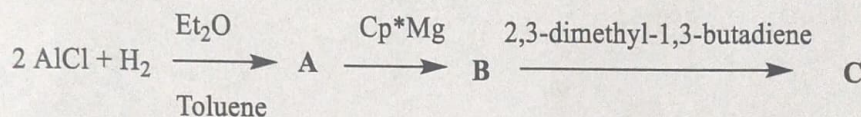
Indian Institute of Science Education and Research (IISER) Mohali

CHM303-Main Group Chemistry; Instructor: Dr. Kuduva R. Vignesh

Second Mid-Semester Exam, (Total = 15 Marks)

PART A (Marks: 5 x 2 = 10)

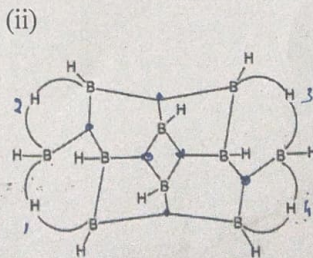
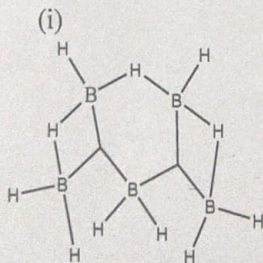
1. Draw structures of structurally related isoelectronic carboranes for Nido- B_6H_{10} , and write the name for each of these?
2. Using orbital interactions between L_nM and NO, explain the formation of linear, $L_nM(\text{linear-NO})$ and bent, $L_nM(\text{bent-NO})$ nitrosyl complexes. Comment on the formal charge and electron contribution of NO in these complexes.
3. Draw the structure of $(R_6B_3N_3)Cr(CO)_6$ as a true π -complex and as a σ -only model and compare this structure with that of $(\eta^6-C_6R_6)Cr(CO)_3$.
4. Find the products A, B and C in the reaction below and draw the structure of B and C, clearly showing the bonding modes.



5. Discuss the bonding situation in Al_2Ph_6 , clearly showing the different bonding modes of the bridging $C_6H_5^-$ anion ($\mu-C_6H_5^-$).

PART B (Marks: 1 x 5 = 5)

6. a) Using the molecular orbital approach, discuss the bonding situation in $[B_6H_6]^{2-}$, clearly mentioning the number and type of B-B bonding, non-bonding and anti-bonding molecular orbitals and the number of electrons occupying them. (2 marks)
- b) Using the polyhedral electron pair approach (Wade's rule), predict the structure of the basic cluster framework and show your calculation of electron counting. (1 mark)
 - (i) $B_2H_7^-$
 - (ii) $(B_5H_8)_2$
- c) Find the STYX number for the following polyhedral boranes and give their nomenclature. (2 marks)



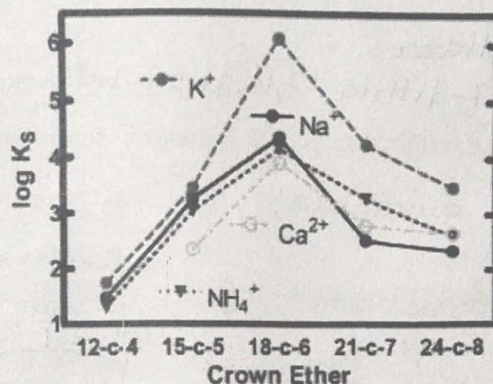
PART A (Marks: 3 x 1 = 3)

(Note: Multiple answers are correct for this part of the questions; unless all the answers are selected correctly in these questions, zero marks will be given. No partial marks for partial answers. Please tick (✓) or circle the answers)

1. Pick the correct statement about Lithium compounds.

- ☒ a) Organolithium compounds have a tendency to form oligomeric units in solution as well as in the solid state.
- ☒ b) A portion of the polymeric structure of LiCp can be found in the lithocene anion $[\text{Li}(\text{C}_5\text{H}_5)_2]^-$
- ☒ c) Structure of the lithocene anion in the crystalline state, staggered rings, D_{5d} symmetry.
- ☐ d) $\text{LiOH} \cdot \text{H}_2\text{O}$ is made up of a double chain

2. Binding constants ($\log K_s$) for 3n-crown-n ($n = 4-8$) compounds with Na^+ , K^+ , Ca^{2+} , and NH_4^+ determined in CH_3OH at 25 °C are shown in the image below.



Choose the correct statement(s) from the graph.

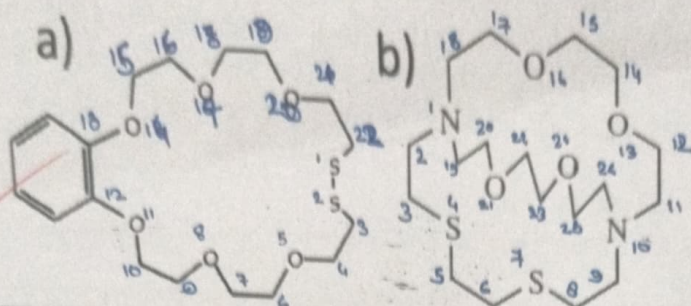
- ☐ a) The selectivity of a crown system is understood based on the hole-size relationship and which can be applied universally.
- ☒ b) The K^+ ion binding was better for all crown ethers
- ☐ c) The K^+ ion binds best to 18-crown-6 but not to other crown ethers
- ☒ d) Na^+ , Ca^{2+} , and NH_4^+ ions were all bound more strongly by 18-crown-6 than by any of the other crown ethers.

3. Pick the incorrect statement about organomagnesium compounds.

- ☐ a) For the preparation of Grignard reagents, the order of reactivity of the CH_3X is as follows: $\text{CH}_3\text{I} < \text{CH}_3\text{Br} < \text{CH}_3\text{Cl}$
- ☐ b) Alkyl/aryl magnesium fluoride reagent is prepared indirectly by using a fluorinating reagent
- ☒ c) Rieke Mg can be obtained when anhydrous MgO_2 is reduced with potassium.
- ☒ d) The solid-phase magnesocene exhibits an eclipsed conformation, but in the gas phase, it exhibits a staggered conformation.

PART B (Answer ALL, Marks: 6 x 2 = 12)

4. Write the IUPAC name of the following compounds



a) ~~1,2-dithia 12-benzo 5,8,11,13-tetraoxa cyclodocosane~~

~~12-benzo 1,2-dithia 5,8,11,14,17,20-tetraoxa cyclodocosane~~

b) ~~4,7-dithia 1,10-diaza 13,16,21,24-tetraoxa bicyclo [8.8.8] hexacosane~~

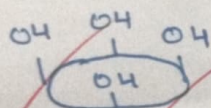
→ ~~4,7-dithia 1,10-diaza~~

1,10-diaza 4,7-dithia 13,16,21,24-tetraoxa [8.8.8] bicyclo hexacosane

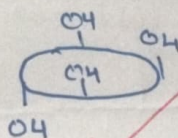
5. Draw conformations of tetrahydroxycalix[4]arenes in schematic form?

→ 4 conformations available;

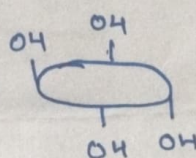
(i) Cone



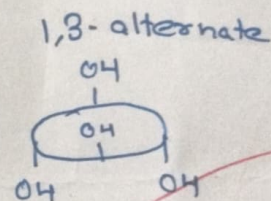
(ii) Partial cone



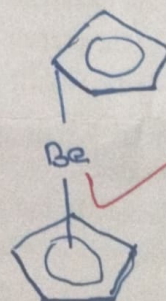
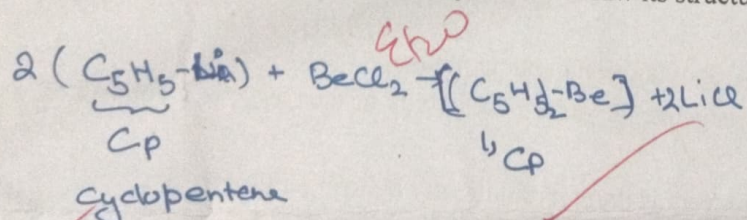
(iii) 1,2-alternate



(iv)

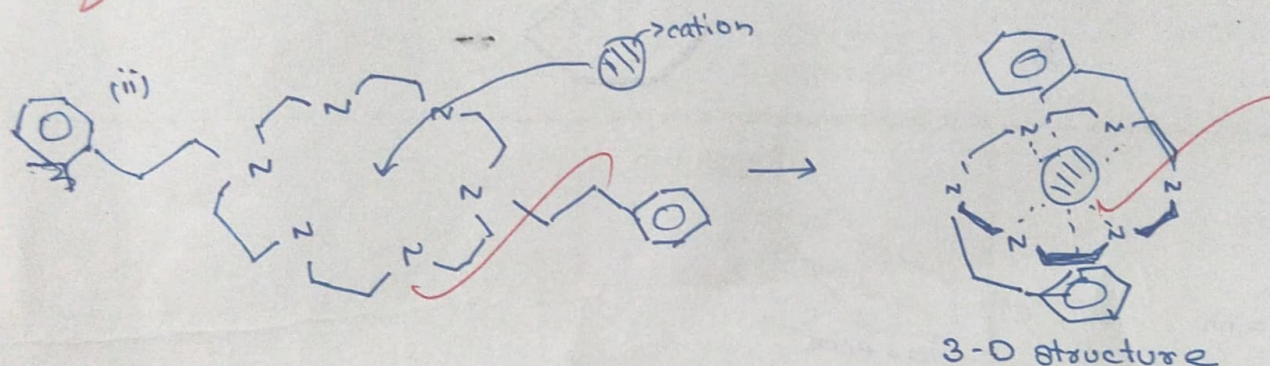
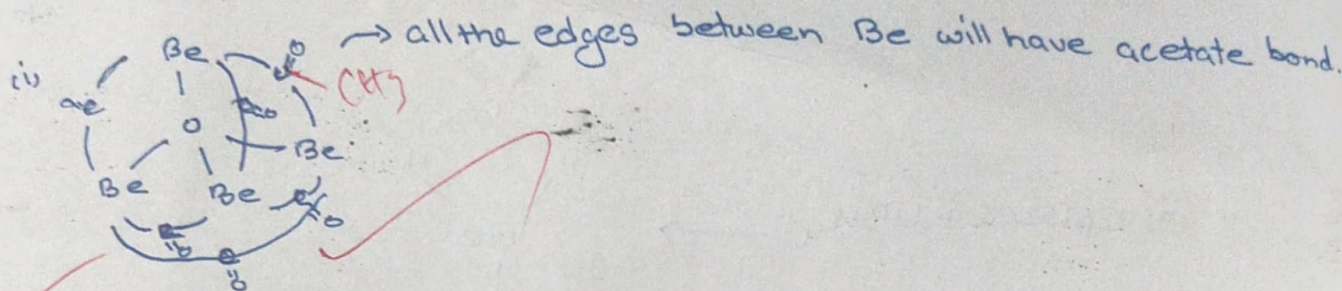


6. How do you synthesise Beryllocene? Draw its structure.



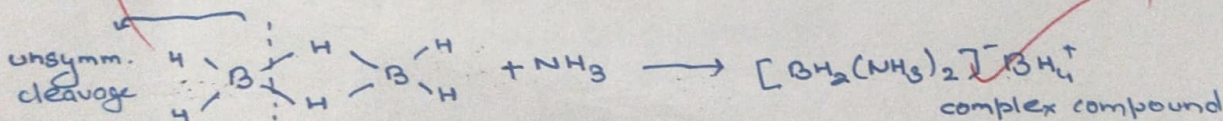
slipped sandwich structure

7. Draw the structure of the following compounds in their stable form: (i) Basic Beryllium acetate, (ii) folded and extended conformations of a lariat ether

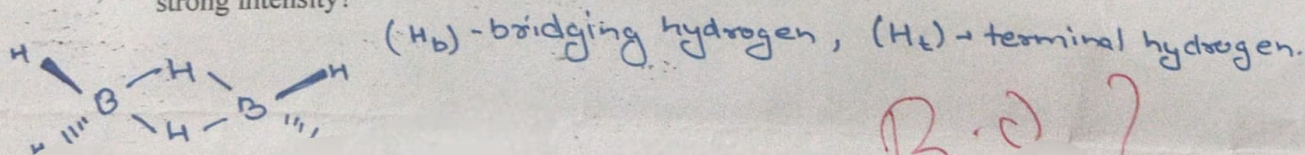


8. i) Does B_2H_6 form adduct with NH_3 ? Justify your answer with the equation for the reaction.

→ No, the direct addition of NH_3 to B_2H_6 does not form an adduct. The reason is because NH_3 is a hard base, hence while bonding NH_3 cleaves the B_2H_6 "asymmetrically" hence forming not an adduct but some other complex.

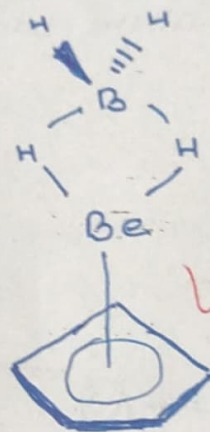
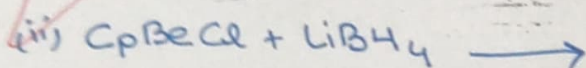
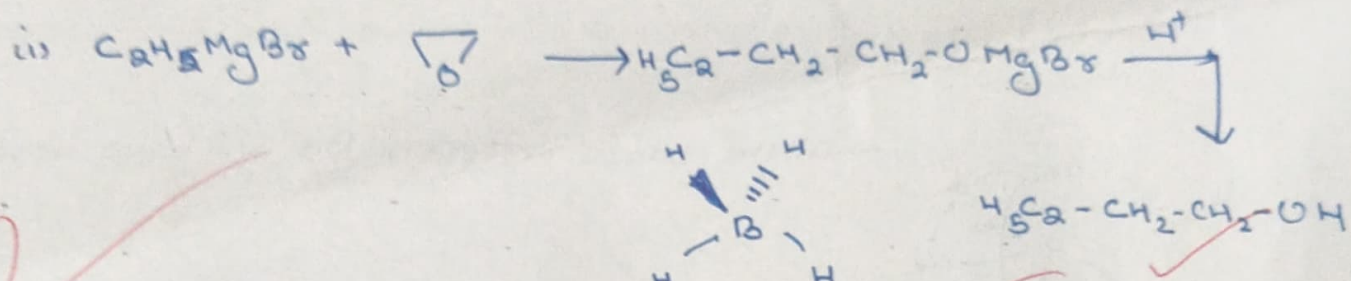


- ii) For diborane, the B-H stretching frequencies for bridging (H_b) are lower compared to the terminal (H_t) hydrogen atoms in IR spectroscopy. Explain why and which one will show a strong intensity?



B.O. ?

9. Write reaction product of (i) C_2H_5MgBr with epoxide; (ii) $CpBeCl$ with $LiBH_4$.



Rough Use

Name: .

Date: 17-09-2025

Reg. No.:

Duration: 30 minutes

CHM303 – Main group chemistry

Quiz-I (Marks: 10 x 1 = 10)

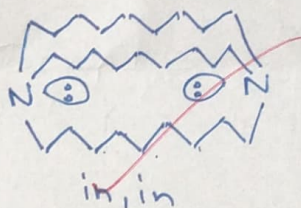
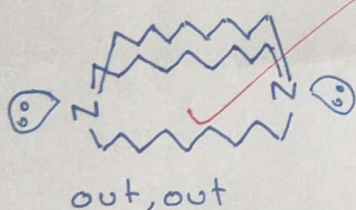
1. Finish the following sentence with the correct statement:

The first short row of elements, Li→Ne, are not representative of the rest of the groups that they head because:

- a) they do not favor the formation of pi bonds
- b) they do not incorporate d-orbitals in their bonding
- c) they adopt similar shapes and geometries to those in the second row

2. Why are a small amount of crystals of I_2 added in the synthesis of Grignard reagents?

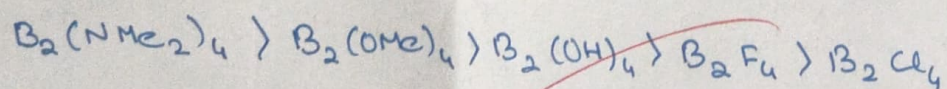
3. Draw the structure of [9.9]amine in both out,out and in,in form.



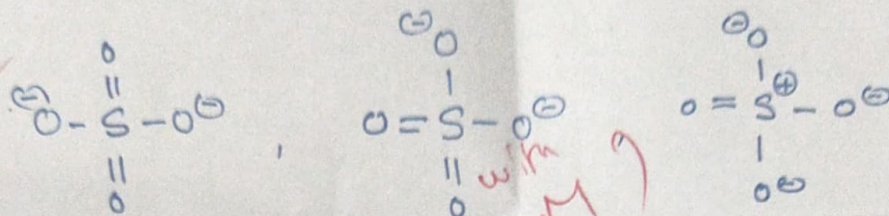
4. Give the exact molecular formula of Borax. Draw its structure.

5. Arrange these compounds based on their thermal stability, considering the expected sequence of $p\pi-p\pi$ bonding between the X group and B:

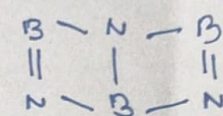
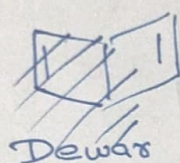
$B_2(OMe)_4$, $B_2(OH)_4$, $B_2(NMe_2)_4$, B_2F_4 , B_2Cl_4



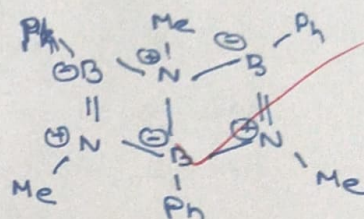
6. Many oxoanions can act as ligands in more than one way (i.e., multiple potential binding modes). Draw three binding modes for SO_4^{2-}



7. Draw the structure of a stable example of Dewar borazine.



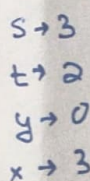
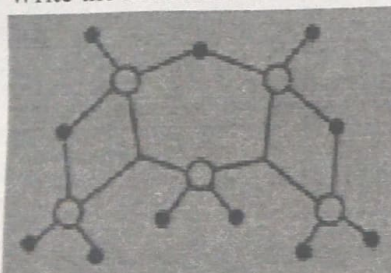
Dewar
borazine
(open book structure)



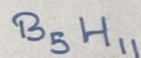
stable example

8. Give an example of Aluminum(I) halides

9. Write the nomenclature for the below borane cluster and find its STYX number.

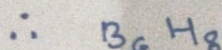
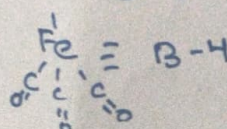
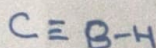
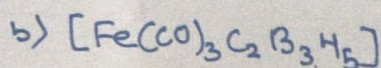
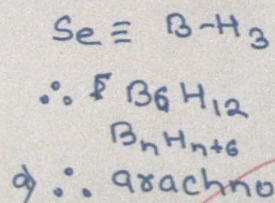
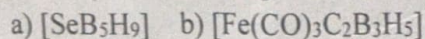


(3203)

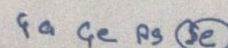
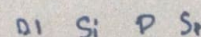
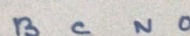
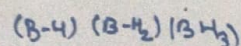


arachno-pentaborane(11)

10. Classify the following metalloborane clusters as closo, nido, or arachno:



b) $\therefore$ closo



Name:

Date: 08-11-2025

Reg. No.:

Duration: 30 minutes

CHM303 – Main group chemistry

Quiz-II (Marks: 5 x 1 = 5)

(Note: Multiple answers are correct for questions 1-4; unless all the correct answers are selected for these questions, zero marks will be awarded. No partial marks for partial answers)

1. Pick the **correct statements** about Boranes and carboranes

- a) The neutral B_6H_8 is a closo-borane. ✓
- b) The neutral $B_5H_9(PMe_3)_2$ is an arachno-borane.
- c) In carboranes, carbon tends to adopt the position of lowest coordination number on the polyhedron and to keep as far from other C atoms as possible.
- d) In nido- and arachno-carboranes, the H_n is more acidic than H_t .

2. Pick the **incorrect statements** about the Organyls of group 13 elements.

- a) The Me_3Ga and Me_3In organyls are dimeric in solution and in the gas phase.
- b) The best-known organometallic compound of Tl^I , $(C_5H_5)Tl$, can even be prepared in an acidic medium.
- c) The Lewis acidity of species R_3M decrease in the order: $B > Al > Ga > In > Tl$.
- d) The Ph_3Al organyl is dimeric in solution and monomeric in the gas phase.

3. Pick the **correct statements** about allotropes of carbon

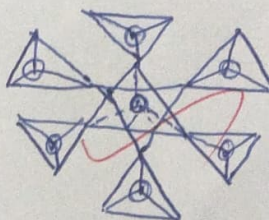
- a) Graphite sheets are 2D sheets held together by weak van der Waals forces, and the carbon atoms are sp^2 hybridised
- b) Diamond is a 3D network of strong covalent bonding between sp^3 hybridised carbon atoms
- c) The C_{60} Fullerene has 60 vertices and 32 faces with 20 pentagons and 12 hexagons.
- d) Carbon nanotubes are 1D sheets of the hexagonal network of carbon rolled to form tubes.

4. Pick the **correct statements** about the Organosilicon compounds.

- a) Organosilyl radicals $R_3Si\cdot$ are trigonal pyramidal.
- b) Organosilyl anions R_3Si^- are tetrahedral.
- c) Ground state for Me_2C is triplet, but Me_2Si , Me_2Ge , and Me_2Sn are singlet.
- d) $R_2Si=SiR_2$ compounds are generally not planar but trans-bent.

5. If silicates are described with a general formula of Si_pO_q , what is the ratio of p:q for Tectosilicates and draw its structure?

$p:q = 1:2$



Tectosilicates